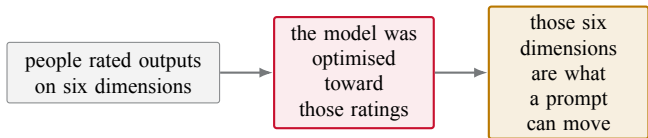




Chapter 3 — Control

The levers that work, and what each one steers

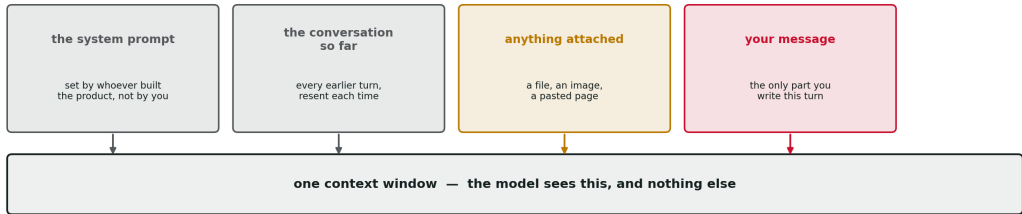
The instructions that steer it are the ones aimed at what was rated



Chapter 2 established the first two boxes. **The third is this chapter**, and it is why the techniques here are not interchangeable with a list of tips: a lever works when it addresses a dimension the rating layer actually scored, and does nothing when it does not.

[--] *Follows from earlier slides in this chapter. No external source.*

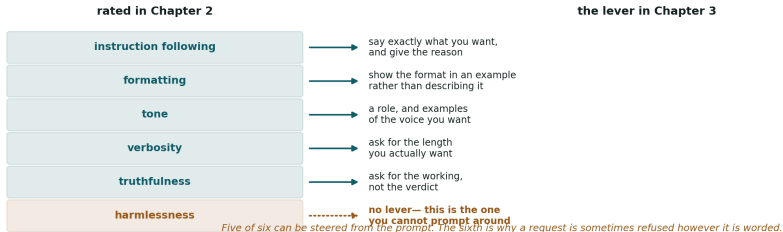
A prompt is four separate things arriving in one window



Chapter 1 defined the **context window**; Chapter 2 defined the **system prompt**. Only the fourth box is written by the person at the keyboard, and three of the four are still there on every turn. Every lever in this chapter is a statement about one of these boxes.

[--] *Follows from earlier slides in this chapter. No external source.*

Every lever in this chapter steers a dimension that was rated



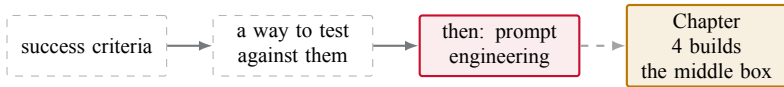
The chapter walks this figure top to bottom. **Harmlessness has no lever**, which is the informative row: a request refused on those grounds is refused however it is worded, and rephrasing to get around it is the one thing on this slide that does not work.

[--] Follows from earlier slides in this chapter. No external source.



Prompt engineering assumes you already have a way to test the result

- ▶ The vendor's own prompting guidance states its prerequisites before any technique: a clear definition of success, a way to measure against it, and prompts to improve.
- ▶ Without the second of those, every change on the following slides is a change whose effect you cannot observe.
- ▶ The failure this produces is specific. A prompt is edited, the next answer reads better, and the improvement is attributed to the edit rather than to sampling.



[V] Anthropic, 2026. *Prompting best practices*, platform.claude.com — vendor documentation, fetched 2026-08-19.

Saying exactly what you want removes the guesses the model would otherwise make

Synthetic example built for this course. Nothing here is quoted from a vendor document.

what gets typed

"Check my concrete mix design."

answerable, but the model must guess the exposure class, the standard, and what you want back

the same request, lever by lever

Check my concrete mix design

- + for a marine tidal-zone pile cap
- + against BS 8500-1
- + flag anything non-compliant and say which clause
- + as a table, one row per parameter

the request

states the exposure class

names the standard to check against

says what the output is for

fixes the format

steers

truthfulness

truthfulness

instruction following

formatting

The weak version is answerable. It is answerable in several incompatible ways, and nothing in it says which one you meant, so the exposure class, the standard and the output format are all chosen for you.

Standing: documented, and each clause maps to a rated dimension.

[V] Anthropic, 2026. *Prompting best practices, platform.claude.com — vendor documentation, fetched 2026-08-19.*

Giving the reason for a constraint makes it cover cases you did not list

constraint only

“Use SI units.”

Applies to the units. Says nothing about what to do with a figure quoted in psi, or a standard that is written in imperial.

constraint with its reason

“Use SI units, because this goes into a report that is checked against Eurocode.”

Now covers the imperial-source case, the mixed-units case, and the question of which standard wins, none of which were listed.

The mechanism is the one Chapter 2 taught: instruction following was rated against the **intent** of an instruction, not its literal surface, so a stated reason gives the model something to generalise from.

Standing: documented, with a stated mechanism.

[V] Anthropic, 2026. *Prompting best practices, platform.claude.com — vendor documentation, fetched 2026-08-19.*

Tags separate the instruction from the data it applies to

run together

```
Summarise this and flag any  
clause that conflicts with  
BS 8500 The contractor shall  
ensure cover of not less than  
...
```

Where the instruction stops and the document starts is
a judgement call, and it is being made for you.

The tag names are arbitrary. What matters is that a boundary exists and that the same one is used consistently within a prompt.

Standing: documented, and this is also the defence Chapter 8 builds on.

separated

```
Summarise the document below  
and flag any clause that  
conflicts with BS 8500.  
<document>  
The contractor shall ensure...  
</document>
```

The boundary is marked rather than inferred.

[V] Anthropic, 2026. *Prompting best practices*, platform.claude.com — vendor documentation, fetched 2026-08-19.

Showing the format works better than describing it

- ▶ A described format has to be interpreted before it can be produced, and an example skips that step.
- ▶ The guidance is specific about quantity and kind: three to five examples, relevant to the task, and diverse across the cases you care about.
- ▶ Diversity is the part usually skipped. Five examples of the same easy case teach the shape of the easy case.



Standing: documented, with stated bounds on how many and what kind.

[V] Anthropic, 2026. *Prompting best practices*, platform.claude.com — vendor documentation, fetched 2026-08-19.

A negative example puts the unwanted pattern into the window

- ▶ The guidance is to say what to do rather than what to avoid, and the reason is mechanical rather than stylistic.
- ▶ Chapter 1 established that everything in the window is scored as context. An example of the output you do not want is still an example of that output.
- ▶ The replacement is not a softer prohibition. It is the positive instruction that makes the prohibition unnecessary.

“Do not write it as a bulleted list.”

“Write it as continuous prose, one paragraph per exposure class.”

Standing: documented.

[V] Anthropic, 2026. *Prompting best practices*, platform.claude.com — vendor documentation, fetched 2026-08-19.

The documentation claims a role changes tone and behaviour

- ▶ What the guidance says a role in the system prompt does: focus **behaviour and tone**. That claim is documented and this course accepts it.
- ▶ What people hear: that telling it to be a structural engineer makes it better at structural engineering. **That is a claim about accuracy**, and it is a different claim.
- ▶ The two are separable, and there is a published result on the accuracy version. It is Chapter 4's, because it needs a measurement to settle rather than an argument.

Standing: the tone claim is documented. The accuracy claim is not established here — Chapter 4.

[V] Anthropic, 2026. *Prompting best practices*, platform.claude.com — vendor documentation, fetched 2026-08-19.

Asking for a length is more reliable than asking for brevity

- ▶ Verbosity was a rated dimension, so it moves. What moves it is a stated target rather than an adjective.
- ▶ “Be concise” is judged against whatever the rating layer settled on as concise. “Two paragraphs, no preamble” is not a judgement call.
- ▶ One version-specific caveat, and it is on the record: on the current model, responses run longer by default than on earlier ones, and the effort setting does not reliably shorten what you see. Prompting for length is the lever that works.

Standing: documented, including the version-specific caveat.

[V] Anthropic, 2026. *Prompting best practices*, platform.claude.com — vendor documentation, fetched 2026-08-19.

The model decides how much to think, and wording biases that decision

the effort dial — not in the chat app

max	always thinks, no limit on depth
xhigh	always thinks deeply
high	almost always thinks (the default)
medium	may skip thinking on simple queries
low	skips thinking where speed matters

Documented for the API and Claude Code.

what you can steer by wording

"Please think hard before responding."

more thinking on this turn

"Answer directly without deliberating."

less thinking on this turn

Suppressing it costs quality on tasks that need it.

Thinking is decided **per request**, by the model, on how hard the request looks. The effort ladder on the left is documented for the API and for the coding tool. In a chat window the lever is the wording, and the cost of suppressing thinking is borne on exactly the tasks that needed it.

[V] Anthropic, 2026. *Steering thinking*, platform.claude.com — vendor documentation, fetched 2026-08-20.

Asking for the working produces claims you can check

you asked for a verdict

"The mix is compliant."

one sentence, nothing in it can be checked without redoing the whole job yourself.

checkable claims: 0

you asked for the working

- w/c ratio 0.45, limit 0.45 for XS3
- cover 45 mm, limit 50 mm — short
- cement type CEM I, permitted
- min strength C35/45, specified C32/40 — short

four claims, each one a number you can look up

checkable claims: 4

Both answers cost about the same to produce. They differ in what they cost **you**: the verdict has to be verified by redoing the work, and the working has four numbers that can each be looked up in a minute.

Standing: derived from the truthfulness dimension. The anti-pattern on the other side of it is frame 19.

[--] Follows from earlier slides in this chapter. No external source.

Attaching the figure beats describing the figure

you describe the plot

"I have a load-displacement curve
with a weird kink about two thirds
of the way along."

the model works from your description,
so it can only be as good as the description

your reading of the plot is the input

you attach the plot



the pixels go into the window
alongside your question

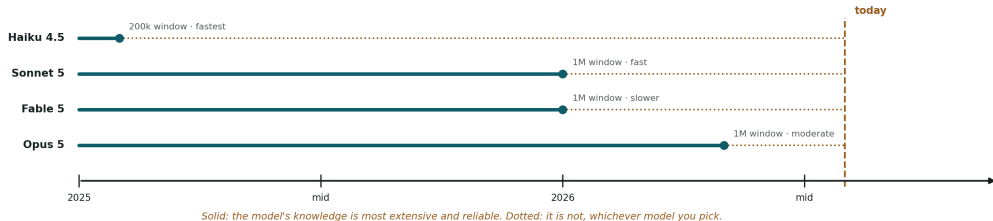
the plot itself is the input

Chapter 1 established that the window takes images as well as text. A described plot is filtered through your reading of it first, so the answer is bounded by the description. An attached plot is not.

Standing: derived from Chapter 1's window definition. File-size limits are Chapter 14's.

[--] Follows from earlier slides in this chapter. No external source.

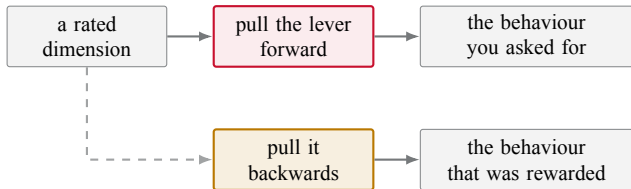
Every model has a date past which the literature is not in it



The picker offers models that differ in speed, in how much they hold at once, and in how recent their knowledge is. For literature work the third column is the one that decides an answer, and **no choice in the list removes the dotted section.**

[V] Anthropic, 2026. *Models overview*, platform.claude.com — vendor documentation, fetched 2026-08-20.

The anti-patterns pull the same levers backwards



Three prompts follow that ask, without meaning to, for exactly the behaviour the rating layer rewarded. **Each one is a lever from earlier in this chapter, reversed**, which is why they are only visible now.

[--] *Follows from earlier slides in this chapter. No external source.*

A question containing its own answer gets that answer agreed with

the answer is visible in the question

“This mix fails because the water/cement ratio is too high, doesn’t it?”

the answer is not

“Here is the mix design and the exposure class. Which parameters, if any, fall outside the limits?”

Agreement with the person asking was rewarded during rating. A question with a preferred answer in it is a request for that answer, and the reply reads as confirmation rather than as a check.

Standing: derived from Chapter 2’s rating mechanism.

[--] *Follows from earlier slides in this chapter. No external source.*

A verdict is cheap to produce and impossible to check

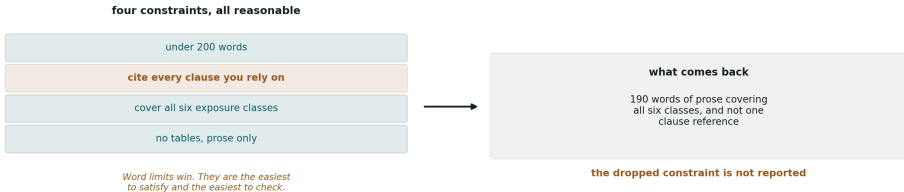
- ▶ Asking “is this right?” asks for a judgement with no working attached, and a confident judgement was the rated outcome.
- ▶ Nothing in a one-line verdict can be verified without redoing the task, so the verdict is either accepted on trust or discarded.
- ▶ The repair is frame 14, applied deliberately: ask for the parameters, the limits and the comparison, and reach the verdict yourself.

Chapter 17 uses the same move as a technique rather than as a repair, on problems where the analysis is the deliverable.

Standing: derived. The reinforcement in Chapter 17 is deliberate.

[--] *Follows from earlier slides in this chapter. No external source.*

Constraints that cannot all hold produce a silent drop



Nothing in the output reports the dropped constraint, and the survivor is generally the one easiest to satisfy and to check. **Ranking the constraints yourself** moves that choice back to you.

Standing: derived from Chapter 2’s conflicting dimensions.

[--] *Follows from earlier slides in this chapter. No external source.*



A prompt tuned on one model is untested on another

- ▶ The prompting guidance is explicitly model-specific, and already carries exceptions for the current model and a deprecation for an older mechanism.
- ▶ A prompt is tuned against one model's rating history. Another vendor's model has a different one, and so does the next version of the same model.
- ▶ The practical form: a prompt that has been working is evidence about the model it was working on, and the same prompt on a new model is an untested change.

Chapter 12 treats portability as its subject. What this chapter needs is the habit of recording which model a prompt was tuned against.

Standing: documented.

[V] Anthropic, 2026. *Prompting best practices*, platform.claude.com — vendor documentation, fetched 2026-08-19.

Five of these are documented, and two are not yet established

What you can act on now

Say what you want; state constraints; give the steps in order

Explain why a constraint exists

Separate instruction from data; 3–5 diverse examples

Say what to do rather than what to avoid

A role changes **tone and behaviour**

Standing

documented; maps to a rated dimension

documented, with a mechanism

documented, with stated bounds

documented

documented

A role improves **accuracy**

“*Think step by step*” helps a current reasoning model

The published chain-of-thought result used eight worked exemplars on a 540-billion-parameter model of its day, which is a different proposition from four words typed at a model that already deliberates.

not established. Chapter 4.

not established. Chapter 4.

[V] Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E., Le, Q. & Zhou, D., 2022. *Chain-of-Thought Prompting Elicits Reasoning in Large Language Models*, arXiv:2201.11903.

What we covered, and what is still open

What we covered

A prompt is four things in one window.

Five of the six rated dimensions have a lever; harmless does not.

A stated reason generalises past the wording it was given in.

Showing a format beats describing one.

Thinking is the model's decision, biased by wording.

The working can be checked; a verdict cannot.

Three anti-patterns, each a lever reversed.

What is still open

Nearly all of the above rests on one vendor's documentation about its own product. Nothing here has been measured by us. **Chapter 4.**

Two specific claims are flagged unsettled: that a role improves accuracy, and that a short reasoning instruction helps a current model.

A prompt that works is evidence about one model on one day. **Chapter 12.**

What happens when the document you pasted contains instructions of its own. **Chapter 8.**

[--] No claim is made on this slide.